Communication as Governance: Measuring Communicative Stance in University Intellectual Property Policies

Working Paper

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Abstract

University intellectual-property policies allocate rights, but they also communicate authority. We develop the Policy Communication Stance Index (PCSI), a human-anchored measure of supportive versus restrictive stance in policy text, built from 87,159 sentences in 518 policies at 150 U.S. research universities, 1944–2025. Written stance leans restrictive: 86 percent of university-years fall below the scale midpoint. Stance is a persistent university attribute, yet observable institutional characteristics explain little of it. It tracks supportive and obligation-oriented wording rather than readability, legal density, or rule form, and it drifts toward restriction over recent decades, within universities, clause topics, and rule forms.

Keywords: institutional governance; policy communication; university technology transfer; intellectual property; text as data; organizational heterogeneity
JEL Codes: O34, O38, I23, C55, K11

1 Introduction

Universities govern faculty inventions through written policy. Intellectual-property policies determine who owns inventions, when they must be disclosed, how rights are assigned, and how licensing revenue is shared, and a large literature studies these provisions and the offices that administer them (Mowery et al., 2001, 2004; Sampat, 2006; Thursby and Thursby, 2002; Jensen et al., 2003; Feldman et al., 2002; Siegel et al., 2003, 2007; Perkmann et al., 2013). Yet even the best-studied provision, the inventor's royalty share, has proven difficult to link to commercialization: estimated incentive effects (Lach and Schankerman, 2008) are sensitive to how policies are coded (Ouellette and Tutt, 2020). Coding the terms of a policy may capture only part of what the policy conveys. The same disclosure or assignment rule can be written as a command or as an offer of assistance. This paper measures that second dimension.

We call it *communicative stance*. Formal texts organize meaning, stabilize expectations, and express relationships of authority within organizations (Smith, 2001; Cooren, 2004; Phillips et al., 2004; Fairhurst and Putnam, 2004; Putnam and Nicotera, 2009; Weick, 1995; Maitlis and Christianson, 2014; Scott, 2014). One policy may present disclosure as categorical compliance under unilateral authority; another may present a comparable requirement as part of a guided process with institutional assistance. The distinction has a precise analogue in the grammar of institutions, which separates the deontic content of a rule (must, may, must not) and its sanction from the actors, aims, and conditions it describes (Crawford and Ostrom, 1995), and in jurisprudence, which distinguishes duty-imposing from power-conferring rules (Hart, 1961). It also matters economically. Controlling signals can reduce the effort of intrinsically motivated agents (Frey, 1997; Bénabou and Tirole, 2003; Falk and Kosfeld, 2006), and law shapes behavior through what it expresses as well as what it sanctions (Sunstein, 1996; McAdams, 2015). The closest existing measurement approach, the survey-based Policy Communication Index of Canary et al. (2013), concerns communication practices and channels. Our object is narrower: the stance encoded in the formal policy text itself.

We construct the Policy Communication Stance Index (PCSI) from intellectual-property policies at 150 U.S. research universities over 1944–2025. The corpus contains 518 source documents and 87,159 cleaned sentences, which yield 480 university-years with policy text and 4,277 annual policy-in-force observations. Human-coded sentences define restrictive, neutral, and supportive communication; a fine-tuned contextual classifier scales those judgments to the full corpus; and calibrated class probabilities are aggregated to the university-year level. The approach follows the text-as-data literature's emphasis on defining the construct before scaling it statistically (Grimmer and Stewart, 2013; Gentzkow et al., 2019; Grimmer et al., 2022; Adcock and Collier, 2001).

The analysis asks three questions. First, how do universities speak? Mean PCSI is 0.437, and 86.0 percent of policy-in-force observations lie below the neutral midpoint of 0.50: restrictive probability mass exceeds supportive mass in most policies. Validation exercises show that the measure tracks the supportive-versus-obligation dimension of policy wording rather than readability, legal-technical density, or the deontic form of the rules a policy states. Rule form

explains under 5 percent of sentence-level variation in stance, and a version of PCSI purged of it correlates 0.94 with the baseline.

Second, is stance changing? Average PCSI declines over the modern period, within fixed university cohorts and in years with source policy text. The decline is not confined to the ownership and assignment clauses revised after *Stanford v. Roche* (2011): it appears within disclosure, revenue-sharing, and administrative language, and within each rule form. Almost none of it reflects a change in which topics policies cover or how many obligations they impose.

Third, who speaks differently? Stance is a persistent institutional attribute: 61.0 percent of variation in the annual panel, and 48.7 percent among years with source text, lies between universities. Private R1 universities write more supportive policies than public R1 universities, a difference of about 0.04 when policies written in the same year are compared, conditional on other characteristics. Land-grant status and other observable characteristics add little, and together they explain only a small share of cross-university differences.

These results make a textual dimension of university governance measurable and show that it is not reducible to the rules it accompanies. Section 2 explains how PCSI is constructed and what it captures. Section 3 reports its level, its evolution over time, and its institutional structure. Section 4 discusses interpretation, mechanisms, and limitations, and Section 5 concludes.

2 Measuring the Policy Communication Stance Index

The measurement task is to convert a qualitative feature of governance—how a university communicates authority, obligation, and support—into a comparable university-year score. We define the policy text and the unit being measured, anchor communicative stance in human sentence judgments, use a contextual classifier to extend those judgments to the full corpus, and aggregate the resulting probabilities into PCSI. We then ask what the index captures.

2.1 Policy Texts and the Unit We Measure

The corpus covers university intellectual-property policies from 150 U.S. research universities between 1944 and 2025: 87,159 cleaned sentences from 518 source documents. These documents govern central parts of the university–inventor relationship, including ownership, disclosure, assignment, licensing authority, commercialization procedures, and revenue distribution.

A university can have more than one relevant policy document in the same year. In those cases we pool all sentences from the contemporaneous documents before constructing a university-year score, which yields 480 university-years with source policy text. Between observed source years, we treat the score of a university's most recent policy as remaining in force until a newer policy source appears. Applying this rule through 2025 produces 4,277 annual policy-in-force university-years. These rows are annual representations of policies in force, not independently collected documents; each retains the source year from which its score is derived, so analyses can always

separate the annual panel from the 480 source-text years. Table 1 summarizes the sample, and Appendix A documents the panel construction.

Table 1: Corpus and Analysis Sample, 1944–2025

Quantity	Value
Universities	150
Analysis period	1944–2025
Source policy documents	518
Cleaned sentences	87,159
University-years with source policy text	480
Annual policy-in-force university-years	4,277

Notes: When several source documents are available for the same university and year, their sentences are pooled before the university-year score is constructed. In the annual panel, that score is carried forward until a newer policy source becomes available.

2.2 Defining Communicative Stance with Human Judgments

We define communicative stance as the supportive, neutral, or restrictive presentation of institutional rights, obligations, authority, and assistance in formal policy text. The construct concerns how governance is expressed. It is not the substantive generosity of the underlying legal rule, generic positive or negative sentiment, or a measured response of faculty readers.

Human annotation gives this distinction operational content. Supportive sentences emphasize assistance, guidance, encouragement, collaboration, shared interests, or facilitation. Restrictive sentences emphasize mandatory assignment, categorical obligations, unilateral authority, prohibitions, sanctions, or compliance. Neutral sentences are primarily definitional, descriptive, or procedural without a clear supportive or restrictive orientation. Coders classify the dominant message of the complete sentence, and disagreements were resolved through human adjudication.

Rule content and communicative stance need not coincide. A disclosure requirement can be framed as categorical compliance or as part of a guided process, and a technically dense sentence can be supportive while a simple one communicates strong institutional control. Table 2 gives examples from the scored corpus; the last two rows state the same disclosure obligation with opposite stance. Appendix B provides the annotation rubric and the reproducibility boundary of the preserved labels.

2.3 Scaling the Human Judgments to the Full Corpus

Human coding defines the classification target but cannot feasibly be applied manually to more than 87,000 sentences. We therefore use supervised natural-language processing as a scaling device. The preserved classifier is a fine-tuned `bert-base-uncased` model (Devlin et al., 2019).

Table 2: Illustrative Sentence-Level Communicative Stance

Class	Illustrative sentence	$\hat{p}_{\text{class}}$
Supportive	"Inventors are encouraged to consult with the OTT concerning their options regarding the release of intellectual property." [Public R1, land-grant, 2012]	0.973
Neutral	"Special procedures relevant to these processes are posted on the website of the Office of the Vice President for Research and Innovation." [Public R1, 2021]	0.980
Restrictive	"As a condition of affiliation with the University, members of the University community are bound by all University policies, including this one." [Private R1, 2005]	0.982
Same disclosure obligation, different framing		
Supportive	"The University intends that the obligation for disclosure shall not unduly interfere with prompt publication of research results." [Public R1, 2010]	0.975
Restrictive	"... Each Invention must be disclosed promptly upon its creation, conception or discovery." [Public R1, 2013]	0.978

Notes: $\hat{p}_{\text{class}}$ is the calibrated model probability assigned to the displayed class. Brackets give the institutional type and source year; institutions are identified in the replication package. The last two rows both state a disclosure obligation. The examples illustrate the annotation construct; they are not an independent accuracy test.

A contextual model is useful because the meaning of terms such as *shall*, *may*, *support*, or *assign* depends on surrounding language.

The preserved labeled sample contains 2,732 sentences: 2,458 for training and 274 held out for validation. On the validation set the classifier achieves accuracy of 0.901 and macro-F1 of 0.894. The model returns a probability distribution over restrictive, neutral, and supportive stance for every sentence, and we apply post-hoc temperature scaling before aggregation (Guo et al., 2017). Appendix B reports the configuration and calibration parameter.

2.4 From Sentence Probabilities to PCSI

Let $(\hat{p}_{Ri}, \hat{p}_{Ni}, \hat{p}_{Si})$ denote the calibrated restrictive, neutral, and supportive probabilities for sentence i. Placing restrictive communication at 0, neutral at 0.5, and supportive at 1, the continuous sentence score is

$$s_i = 0.5\,\hat{p}_{Ni} + \hat{p}_{Si}. \tag{1}$$

A sentence with probability concentrated on the supportive class receives a value near 1, one concentrated on the restrictive class a value near 0, and one concentrated on the neutral class a value near 0.5. For university j in year t, baseline PCSI is the unweighted mean of sentence scores in the pooled policy text,

$$\text{PCSI}_{jt} = \frac{1}{N_{jt}} \sum_{i=1}^{N_{jt}} s_i = 0.5 + 0.5\,(\bar{\hat{p}}_S - \bar{\hat{p}}_R)_{jt}\,. \tag{2}$$

The range is $[0, 1]$. Values below 0.5 indicate that restrictive probability mass exceeds supportive mass on average across the policy. A value of 0.5 does not require every sentence to be neutral; it can also result from offsetting supportive and restrictive mass.

2.5 What the Measure Captures

Before interpreting PCSI substantively, we ask whether it depends on how sentence scores are combined, which recognizable features of policy language it reflects, and whether it merely restates the form of the rules a policy contains.

Aggregation. We reconstruct the policy score using the within-policy median, a 5-percent trimmed mean, and a sentence-length-weighted mean. The trimmed and length-weighted versions remain very close to the baseline ($r = 0.999$ and 0.930), while the median preserves much of the ranking (Spearman $\rho = 0.878$) but changes cardinal distances more noticeably (Appendix C).

Linguistic content. We construct three transparent diagnostics from the same text. *Tone* summarizes supportive and collaborative language relative to obligation-, control-, and sanction-oriented wording. *Clarity* summarizes sentence structure and procedural guidance. *Legal Load* summarizes legal and IP-technical terminology (Appendix E). The diagnostics are not alternative definitions of stance and do not enter PCSI. If PCSI captures the construct defined above, it should be related most strongly to Tone and should not reproduce readability or legal density.

Table 3 confirms this pattern. Entered alone, Tone accounts for 21.6 percent of in-sample variation in PCSI. Adding Clarity leaves the fit unchanged, and adding Legal Load raises R^2 to 0.236. Because nested R^2 comparisons depend on the order of entry, the table also reports Shapley shares, which average each diagnostic's marginal contribution over all orderings (Lindeman et al., 1980): Tone accounts for 89 percent of the explained variation, Legal Load for 9 percent, and Clarity for 2 percent. Using only the 480 source-text university-years, Tone's share is 78 percent. Tone and Legal Load are nearly uncorrelated ($r = 0.029$), so a policy can be technically dense while communicating assistance, and a simple policy can communicate categorical control. The three diagnostics together explain about a quarter of PCSI; the remainder may reflect combinations of language that lexical measures miss, such as discretion, relational framing, or the sequencing of obligations and assistance, as well as measurement error.

Rule form versus framing. A stronger concern is that PCSI simply counts rules: a policy full of obligations would score as restrictive regardless of how it states them. We code each sentence's deontic operator following the Institutional Grammar of Crawford and Ostrom (1995)—obligation (*shall*, *must*, *is required*), prohibition (*shall not*, *may not*), permission (*may*, *is entitled*), or none—and flag sanction language, the grammar's "or else" component. Obligations are the most restrictive form on average (mean sentence score 0.329, against 0.467 for sentences without a deontic operator), but the variation within forms dwarfs the differences between them: the eight

Table 3: Interpreting PCSI with Transparent Linguistic Features

	(1)	(2)	(3)
Tone Index	0.064***	0.063***	0.062***
	(0.012)	(0.012)	(0.012)
Clarity Index		0.001	0.006
		(0.008)	(0.008)
Legal Load Index			0.013**
			(0.006)
R^2	0.216	0.216	0.236
Shapley share of R^2: Tone / Clarity / Legal Load			0.89 / 0.02 / 0.09
Same, years with source text			0.78 / 0.14 / 0.09
Observations	4,277	4,277	4,277

Notes: Dependent variable is policy-in-force PCSI; 150 universities. Standard errors clustered by university are in parentheses. Shapley shares average each index's marginal contribution to R^2 over all orders of entry and sum to one. Each diagnostic combines standardized components, so one unit of a composite should not be read as one standard deviation of that composite. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$.

form-by-sanction cells explain only 4.8 percent of sentence-level variance in stance (Appendix Table G2). Prohibitions are no more restrictive than permissions on average, a reminder that a sentence's deontic operator says little about how it is framed. Subtracting each sentence's cell mean before aggregation yields a measure of framing net of rule form that correlates 0.94 with baseline PCSI and reproduces every headline result below (Appendix H). PCSI therefore measures how rules are stated, not how many of each kind a policy contains.

These exercises define the scope of the measure. PCSI is a property of formal policy text. It is not a measure of substantive legal generosity, faculty perceptions, actual technology-transfer practice, or commercialization performance.

3 The Empirical Structure of Communicative Stance

We now describe the measure's level, its evolution over time, and its distribution across universities. The analyses are descriptive; none treats institutional characteristics or policy language as causes.

3.1 PCSI Lies Below the Neutral Midpoint

Figure 1 plots PCSI across the 4,277 policy-in-force university-years, and Table 4 reports summary statistics. Mean PCSI is 0.437, the median is 0.427, and the standard deviation is 0.066. The interquartile range runs from 0.394 to 0.471, with observed values from 0.264 to 0.730.

The central distributional result is the location of the mass relative to the scale midpoint: 86.0 percent of annual observations lie below 0.50. Because PCSI equals $0.5 + 0.5(\bar{\tilde{p}}_S - \bar{\tilde{p}}_R)$, a value below 0.50 means that restrictive probability mass exceeds supportive mass on average across the policy. It does not mean that every sentence is restrictive or that the underlying rules

are unfavorable to inventors. The scale also contains meaningful dispersion: a one-standard-deviation difference in PCSI corresponds to a 0.132 difference in the average supportive-minus-restrictive margin, and the right tail extends to 0.730, showing that substantially more facilitative communication is possible within the same policy genre.

The pattern does not arise from carrying scores forward. Among the 480 university-years with source policy text, mean PCSI is 0.429 and 89.6 percent fall below 0.50. The midpoint is a scale reference, not a normative benchmark.

Table 4: Distribution of PCSI

	Policy-in-force panel	Years with source text
Mean	0.437	0.429
Median	0.427	0.423
Standard deviation	0.066	0.063
25th percentile	0.394	0.391
75th percentile	0.471	0.464
Minimum	0.264	0.264
Maximum	0.730	0.730
Share below 0.50	86.0%	89.6%
Observations	4,277	480

Notes: The policy-in-force panel carries each university's most recent score forward until a newer policy source appears; the second column uses only university-years for which policy text is directly available. Analysis window 1944–2025; 150 universities.

3.2 PCSI Drifts Toward Restriction Over the Modern Period

Figure 2 plots annual mean PCSI beginning in 1978, the first year in which at least five universities are represented. Average stance is highest around 1980, at roughly 0.49, and declines over subsequent decades to approximately 0.421 in 2025.

The raw series combines changes in policy language with changes in which universities the archive covers. Within fixed university cohorts the decline persists. Among the 93 universities already represented by 2000, mean PCSI falls from 0.447 to 0.422 by 2025, and the university fixed-effects trend is -0.00112 per year (clustered SE $= 0.00042$, $p = 0.008$). Cohorts beginning in 1990, 1995, and 2005 show the same pattern (Appendix Table F1). Removing carry-forward and using only the 467 source-text university-years dated 1980 or later gives a fixed-effects trend of -0.00117 (SE $= 0.00039$, $p = 0.003$); starting in 1990 gives a similar estimate, while the post-2000 estimate is negative but less precise (Appendix Table F2).

Is the decline a change in rules or in stance? One explanation for the drift is substantive. After *Stanford v. Roche* (2011), many universities rewrote assignment provisions as present assignments ("hereby assigns"), and ownership language is the most restrictive part of a typical policy. If

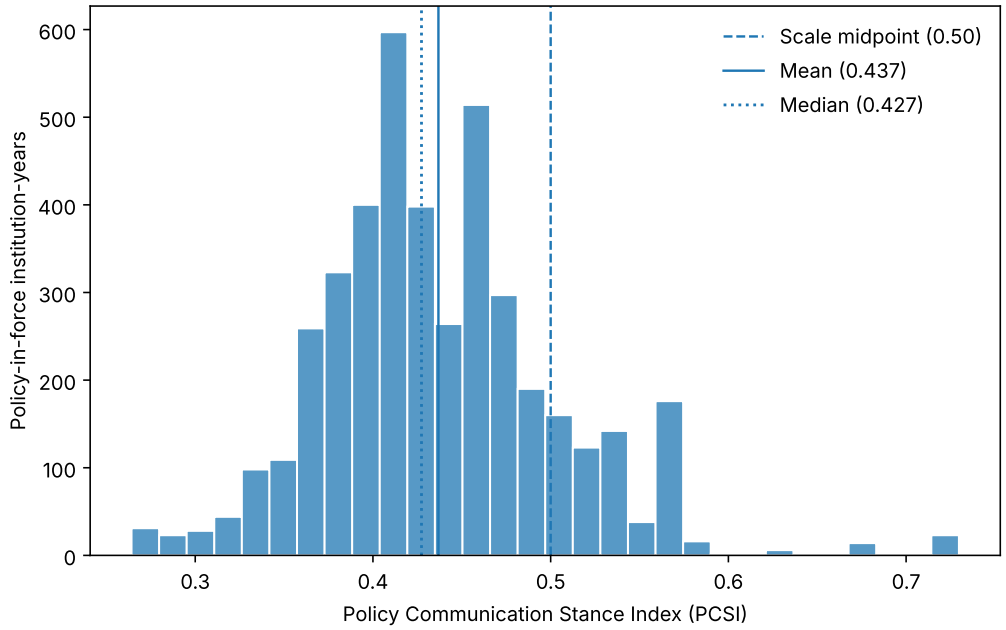


Figure 1: Distribution of PCSI in the 1944–2025 Policy-in-Force Panel ($N = 4{,}277$). The dashed line marks the scale midpoint of 0.50; the solid and dotted lines mark the mean and median.

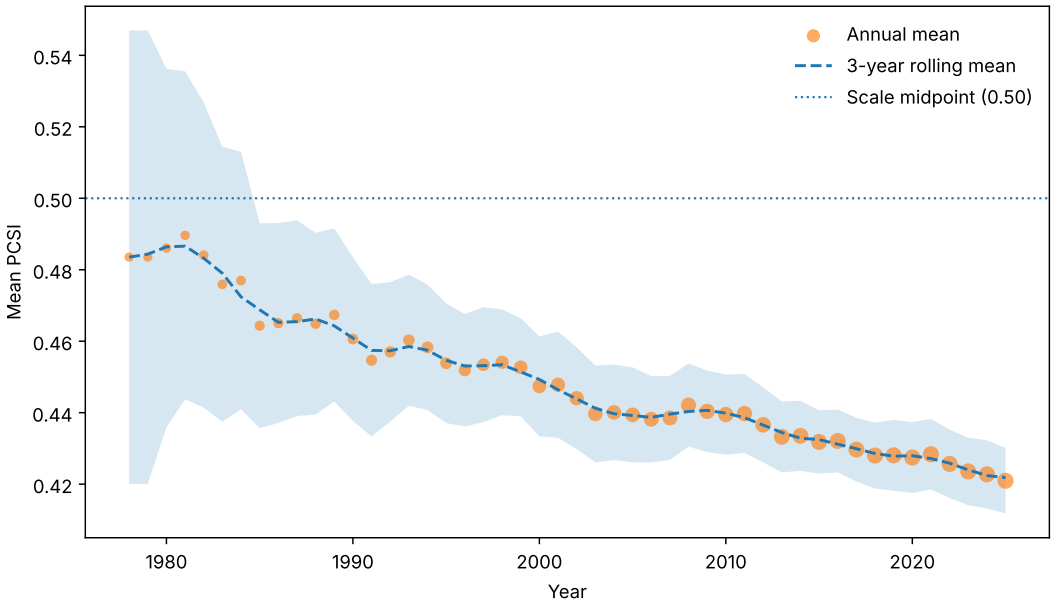


Figure 2: Annual Mean PCSI, 1978–2025. Points show annual cross-sectional means in the policy-in-force panel, with point size proportional to the number of universities represented. The shaded band is the 95% confidence interval for the annual mean, the dashed line is the three-year rolling mean, and the dotted horizontal line marks the scale midpoint of 0.50. Years before 1978 are omitted because fewer than five universities are represented.

the decline reflected such rule changes, it would be concentrated in ownership and assignment sentences, or in a rising share of them, or in a rising share of obligations. We test each possibility.

First, we assign sentences to clause topics with transparent keyword rules—ownership and assignment, disclosure, revenue sharing, licensing and commercialization, sponsored research, copyright and software, administration and disputes, and definitions (Appendix G)—and estimate university fixed-effects trends in mean sentence score within each topic, using source-text years from 1980. Figure 3 shows that the decline is broad. The trend in ownership and assignment language is close to zero (-0.0003 per year, $p = 0.64$), while disclosure (-0.0022, $p = 0.04$), revenue sharing (-0.0017, $p = 0.01$), and administrative language (-0.0013, $p = 0.02$) move toward restriction, and every other topic has a negative point estimate.

Second, a shift-share decomposition for the 63 universities with source text in both 1980–1999 and 2010–2025 separates change within topics from change in the mix of topics. Of a decline of about 0.031 in mean sentence score, -0.028 occurs within topics and -0.003 reflects composition. Ownership language does take up a larger share of later policies (21 percent of sentences, against 17 percent), but this composition effect is small. The same decomposition across rule forms attributes -0.030 to change within forms and $+0.002$ to the mix of forms: later policies do not impose proportionally more obligations, but they state obligations, permissions, and descriptive provisions alike in more restrictive terms (Appendix Table G2).

The modern decline is therefore a change in how policies speak across the board, not only in what assignment clauses require.

3.3 Stance Is a Persistent Institutional Attribute, Weakly Explained by Observables

Where the variation lies. In the annual policy-in-force panel, 61.0 percent of the total sum of squared deviations in PCSI is between universities and 39.0 percent is within universities over time. Because the panel repeats a score until a newer source appears, carry-forward mechanically increases persistence; using only the 480 source-text university-years lowers the between-university share to 48.7 percent (Appendix Table C2). Carry-forward matters for the magnitude, but it does not create the cross-university structure. Revisions are typically incremental: across 330 consecutive policy revisions within universities, the median absolute change in PCSI is 0.021. Stance behaves like an institutional style, a persistent trait of each university's governance text (compare Bertrand and Schoar, 2003).

Who speaks differently. We next ask whether observable characteristics account for these differences.[1] Table 5 reports university-level means, where each university contributes the average of its annual scores so that institutions with longer archives do not receive more weight. We use

[1] R1 and R2 denote doctoral universities with very high and high research activity in the Carnegie Classification of Institutions of Higher Education; control distinguishes public from private universities. Land-grant universities are those designated under the Morrill Acts of 1862 and 1890. Medical-school status indicates whether the university operates a medical school. Urbanicity classifies each campus as rural, semi-urban, or city. Region groups states into six U.S. regions. Classifications are time-invariant attributes recorded in the replication data.

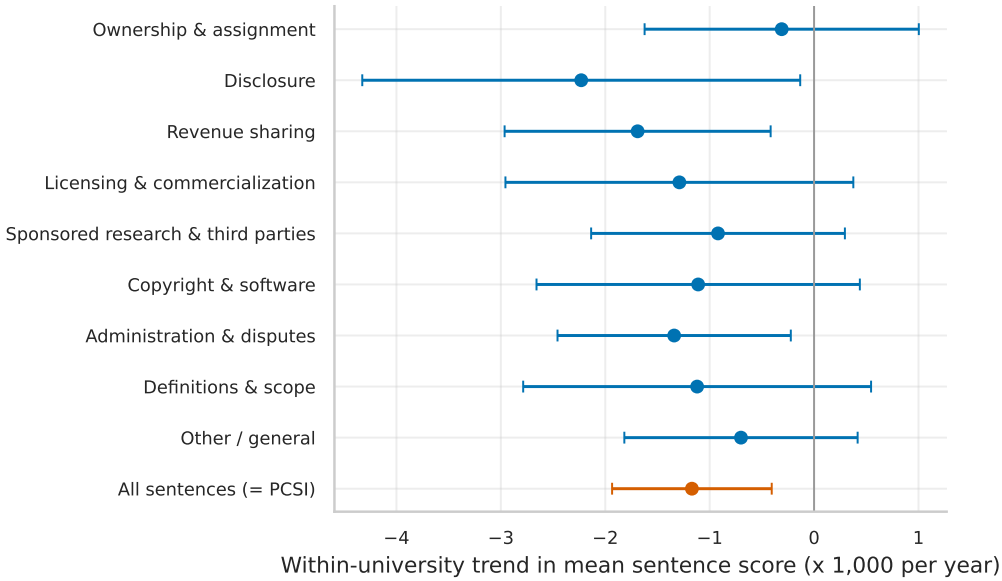


Figure 3: Within-University Trends in Sentence Stance by Clause Topic, 1980–2025. Each point is the coefficient on year from a regression of the university-year mean sentence score within a topic on year and university fixed effects, using source-text university-years from 1980 onward and universities observed in at least two such years; coefficients are multiplied by 1,000. Bars are 95% confidence intervals with university-clustered standard errors. Topics are assigned by keyword rules and can overlap (Appendix G).

rank-based tests because several groups are small (eight private R2 universities) and university means are skewed. Private R1 universities average 0.456, compared with 0.427 among public R1 universities (Kruskal–Wallis $p = 0.004$ across the four types). Land-grant universities average 0.420, compared with 0.442 among other institutions (Wilcoxon rank-sum $p = 0.010$). Both differences survive a Holm adjustment across the four comparisons; medical-school status and urbanicity do not (Appendix Table D1). Both directions hold under the alternative aggregations and when university means use only source-text years (Appendix D).

Table 6 considers the characteristics jointly. Columns (1)–(3) use university means. Relative to public R1 universities, the private-R1 coefficient is 0.029 (SE 0.009) with type alone, 0.019 with land-grant, medical-school, and urbanicity controls, and 0.020 (SE 0.014) with region fixed effects. The attenuation is not a sample artifact: Column (1) is unchanged at 0.029 when estimated on the 144 universities with complete covariates.

University means, however, mix institutional differences with the decades in which each university's policies were written, and with 144 observations they are imprecise. Column (4) instead uses the source-text university-years, includes source-year fixed effects so that each policy is compared with policies written in the same year, and clusters standard errors by university. In this specification the private-R1 difference is 0.040 (SE 0.012) with all controls and region fixed effects, while the land-grant coefficient is -0.005 and imprecise. Weighting university means by the number of source policy years or sentences, or removing calendar-year means before averaging,

Table 5: University-Level PCSI by Institutional Characteristics

Group	*N*	Mean	SD	Range
Carnegie classification and control				
Private R1	38	0.456	0.046	[0.337, 0.534]
Public R1	86	0.427	0.051	[0.328, 0.585]
Private R2	8	0.406	0.053	[0.342, 0.459]
Public R2	18	0.441	0.058	[0.376, 0.580]
Mission and program				
Land-grant	47	0.420	0.047	[0.338, 0.585]
Non-land-grant	103	0.442	0.053	[0.328, 0.580]
Medical school	118	0.438	0.053	[0.328, 0.585]
No medical school	32	0.422	0.047	[0.342, 0.519]
Urbanicity				
Rural	15	0.425	0.041	[0.372, 0.519]
Semi-urban	48	0.422	0.047	[0.338, 0.539]
City	82	0.443	0.057	[0.328, 0.585]

Notes: Unit of analysis is the university; PCSI is averaged across each university's policy-in-force years before group statistics are computed. Five universities lack urbanicity data. Rank-based tests are reported in Appendix D.

leaves the qualitative pattern unchanged (Appendix Table D2).

The private-R1 difference is therefore the one robust institutional regularity; the land-grant difference largely reflects overlap with other characteristics. Even so, observable characteristics explain at most about a sixth of cross-university variation in average PCSI. Broad institutional categories organize some communicative variation without determining it.

4 Discussion

This paper operationalizes communicative stance in written university intellectual-property policy. Its contribution is a measurement framework and a set of descriptive regularities, not evidence that changing policy language changes technology-transfer outcomes.

Why stance might matter. Three pathways make the construct worth measuring. First, faculty and other researchers may encounter policy language directly and use it to interpret obligations, discretion, and available assistance. If they do, theories of motivation crowding-out predict that controlling language can reduce voluntary disclosure by intrinsically motivated researchers (Falk and Kosfeld, 2006; Bénabou and Tirole, 2003), and theories of procedural justice predict that respectful framing supports voluntary compliance (Tyler, 1990)—a live concern given that invention disclosure is widely regarded as incomplete (Jensen et al., 2003). That channel requires exposure and comprehension, neither of which is observed here. Second, technology-transfer staff may reproduce formal policy language in forms, training, and interactions with researchers, so

Table 6: Institutional Correlates of PCSI

	University means			Source-text years
	(1)	(2)	(3)	(4)
Private R1	0.029***	0.019	0.021	0.040***
	(0.009)	(0.012)	(0.014)	(0.012)
Private R2	−0.021	−0.031	−0.035	−0.006
	(0.021)	(0.023)	(0.023)	(0.018)
Public R2	0.014	0.007	0.011	0.005
	(0.015)	(0.020)	(0.018)	(0.011)
Land-grant		−0.014	−0.011	−0.005
		(0.012)	(0.013)	(0.012)
Medical school		0.014	0.010	0.006
		(0.011)	(0.011)	(0.009)
Urbanicity controls	No	Yes	Yes	Yes
Region fixed effects	No	No	Yes	Yes
Source-year fixed effects	No	No	No	Yes
R^2	0.074	0.111	0.161	0.291
Observations	150	145	144	452
Universities	150	145	144	144

Notes: Dependent variable is PCSI. Columns (1)–(3) use one observation per university (mean policy-in-force PCSI) with HC3 robust standard errors; Columns (2)–(3) omit universities with missing covariates. Column (4) uses university-years with directly available policy text, compares policies written in the same year through source-year fixed effects, and clusters standard errors by university. Public R1, rural, and Midwest are the omitted categories. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$.

textual stance could shape implementation even when faculty do not read the original document. Third, policy language may reflect broader organizational culture or leadership priorities (Putnam and Nicotera, 2009; Maitlis and Christianson, 2014; Fairhurst and Putnam, 2004). The current data do not distinguish among these pathways.

Why stance persists and drifts. Persistence combined with weak observable correlates points to institution-specific drafting histories. Policy text is often copied from templates and revised incrementally by counsel, so wording can persist for reasons unrelated to current rules (Kahan and Klausner, 1997; Choi et al., 2013). The legal professionalization of compliance offers one account of the modern drift toward restriction (Edelman, 2016); the evidence that the drift appears within clause topics and rule forms, rather than through new obligations or reworded assignment clauses alone, is consistent with a change in drafting register rather than in substantive rules. The institutional results fit the broader technology-transfer literature, which emphasizes heterogeneous professional routines, faculty–TTO relationships, and local commercialization practices (Owen-Smith and Powell, 2001; Siegel et al., 2003; Markman et al., 2005; Bercovitz and Feldman, 2008; Siegel et al., 2007; Rothaermel et al., 2007; Perkmann et al., 2013). PCSI adds a measurable textual dimension to that heterogeneity.

Limitations. The annual panel assumes that the most recently available policy remains in force until a newer source is observed. Analyses restricted to years with source text reduce reliance on that assumption and yield the same central patterns, but they cannot establish that the historical archive is complete. Sentence-level classification omits layout, headings, sequencing, and long-range relationships among passages, and the keyword rules used to assign topics and rule forms are transparent but coarse. Downstream regressions use predicted rather than observed labels; the headline results are unchanged when PCSI is built from hard labels or from the two-thirds of sentences the model classifies with probability of at least 0.9, and model confidence is stable across decades (Appendix H), but formal corrections for generated variables (Egami et al., 2023; Battaglia et al., 2024) require gold-standard labels linked to the analysis sample. Most fundamentally, textual stance can diverge from substantive legal rights and from actual implementation. A higher PCSI should not be interpreted as a welfare ranking, a measure of legal generosity, or a recommendation to weaken legitimate disclosure and ownership requirements.

Reproducibility. The replication package reproduces saved-model inference, calibrated sentence scores, university-year aggregation, panel construction, and every table and figure in this paper. The original fine-tuning run cannot presently be reconstructed from coder-level pre-adjudication records, so we report only annotation and validation quantities supported by the preserved artifacts. A fresh double-coding of a random sample of sentences would allow inter-rater reliability to be reported for the construct independently of the lost records.

5 Conclusion

This paper makes the communicative stance of university intellectual-property policy measurable. PCSI anchors supportive and restrictive language in human judgment, scales it with a calibrated classifier, and yields a panel covering 150 research universities over eight decades. The measure is a property of text, not of rights, practice, or outcomes, and its value lies in the questions it allows researchers to ask next. Does restrictive framing reduce invention disclosure, as theories of motivation crowding-out predict? Survey experiments that vary policy wording while holding rules fixed could answer that directly. Does stance spread through shared templates and legal counsel, and does it respond to legal events such as *Stanford v. Roche*? Does textual stance carry into technology-transfer practice, or remain on paper? Linking PCSI to licensing, startup, and faculty-survey data is the natural next step.

Data and Code Availability

The data and replication materials used to construct PCSI and reproduce every table and figure are publicly available at https://github.com/krish533/Tech-transfer-1/tree/main/P1_replication_package.

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A Data Construction and Sample Provenance

The analysis uses policy documents from 150 U.S. research universities over 1944–2025: 87,159 cleaned sentences from 518 source documents. When more than one relevant document is available for the same university and year, the sentences are pooled before the university-year score is constructed, which yields 480 university-years with source policy text. The annual policy-in-force panel treats the most recently available score as remaining in force until a newer policy source appears for the same university; carrying scores forward through 2025 produces 4,277 annual observations. Each annual row retains the source year from which its score is derived. For cross-sectional comparisons, annual scores are averaged within university so that every university contributes one observation. The underlying archive also contains an isolated one-sentence 1925 record, retained for provenance but excluded from the analysis window.

B Annotation, Model, and Replication Provenance

Annotation rubric. Sentences are classified as supportive, neutral, or restrictive. Supportive language emphasizes assistance, guidance, encouragement, collaboration, shared interests, or facilitation. Restrictive language emphasizes mandatory assignment, unilateral institutional authority, prohibitions, sanctions, forfeiture, or categorical compliance obligations. Neutral language is primarily definitional, descriptive, or procedural without a clear directional stance. The coding rule concerns the dominant communicative stance of the complete sentence. Multiple trained human coders applied the rubric, and disagreements were resolved through human adjudication. The preserved artifacts contain 2,732 adjudicated labeled sentences. The coder-level pre-adjudication file needed to reproduce inter-rater reliability statistics is not preserved, so numerical reliability measures are not reported.

Classifier and calibration. Table B1 reports the configuration of the preserved classifier. Post-hoc temperature scaling is applied before probability aggregation. The replication package reproduces inference from the saved classifier, calibrated sentence scores, aggregation, panel construction, and the manuscript analyses; the original fine-tuning run cannot presently be reconstructed.

C Measurement Robustness

Table C1 compares the baseline with the within-policy median, a 5-percent trimmed mean, and a sentence-length-weighted mean. Trimming changes almost nothing, length weighting remains strongly aligned with the baseline, and median aggregation produces the largest cardinal difference while retaining a Spearman correlation of 0.878. Table C2 repeats the between/within decomposition under each aggregation and in the 480 source-text years. Carry-forward increases measured persistence under every construction, but substantial cross-university structure remains

Table B1: Preserved Classifier Configuration

Parameter	Value
Base model	`bert-base-uncased`
Transformer layers	12
Attention heads	12
Hidden size	768
Intermediate size	3,072
Maximum input length	256 tokens
Output classes	3
Training sentences	2,458
Validation sentences	274
Total labeled sentences	2,732
Temperature parameter	$T = 0.8462$
Validation accuracy	0.901
Validation macro-F1	0.894

Notes: Configuration and metrics are read from the preserved model artifacts in the replication package.

without it.

Table C1: Sensitivity to Alternative Sentence Aggregation Rules

Aggregation	Mean	Pearson r	Spearman ρ	Midpoint agreement
Baseline (unweighted mean)	0.437	1.000	1.000	1.000
Median PCSI	0.471	0.623	0.878	0.937
5% trimmed mean	0.431	0.999	1.000	1.000
Length-weighted mean	0.433	0.930	0.906	0.933

Notes: $N = 4{,}277$ policy-in-force university-years. Correlations are with baseline PCSI. Midpoint agreement is the share of observations that the baseline and the alternative place on the same side of 0.50.

D Institutional Tests

Table D1 reports the four rank-based institutional comparisons with Holm-adjusted p-values. The unconditional patterns are stable across aggregation choices: the private-R1 minus public-R1 difference is 0.029 with baseline PCSI, 0.018 with the median, and 0.032 with both the trimmed and the length-weighted mean; the land-grant minus non-land-grant difference ranges from -0.020 to -0.024. When university averages use only source-text years, the two differences are 0.033 and -0.019.

Table D2 reports the private-R1 and land-grant coefficients across samples, weights, and era adjustments. Archive vintage differs modestly by type: the mean source-policy year is 2002.3 for private R1 universities and 2005.1 for public R1 universities. Because PCSI declines over time, comparing university means without adjusting for era slightly favors universities with older

Table C2: Between- and Within-University Variation

Specification	Between share	Within share
Policy-in-force panel		
Baseline	0.610	0.390
Median	0.434	0.566
Trimmed mean	0.616	0.384
Length-weighted mean	0.589	0.411
Years with source text		
Baseline	0.487	0.513
Median	0.300	0.700
Trimmed mean	0.494	0.506
Length-weighted mean	0.463	0.537

Notes: Shares partition the total sum of squared deviations of PCSI into between- and within-university components.

Table D1: Institutional Comparisons with Multiple-Testing Adjustment

Characteristic	Test	Raw p	Holm p
Carnegie classification and control	Kruskal–Wallis	0.004	0.014
Land-grant status	Wilcoxon rank-sum	0.010	0.029
Urbanicity	Kruskal–Wallis	0.055	0.111
Medical-school status	Wilcoxon rank-sum	0.172	0.172

Notes: University-level mean PCSI ($N = 150$; 145 for urbanicity). The Holm adjustment is applied to this family of four tests.

archives, which is why Column (4) of Table 6 compares policies within source year. The private-R1 difference is larger, not smaller, in that comparison.

Table D2: Private-R1 and Land-Grant Coefficients Across Samples and Estimators

	Type only	Full controls		
	Private R1	Private R1	Land-grant	N
University means	0.029***	0.021	−0.011	144
	(0.009)	(0.014)	(0.013)	
University means, common sample	0.029***	0.021	−0.011	144
	(0.009)	(0.014)	(0.013)	
WLS, weight = source policy years	0.027**	0.020	−0.021	144
	(0.010)	(0.015)	(0.015)	
WLS, weight = sentences	0.021	0.023	0.005	144
	(0.014)	(0.023)	(0.028)	
Era-adjusted university means	0.029***	0.022	−0.009	144
	(0.009)	(0.014)	(0.013)	
Source-text years, year FE	0.040***	0.040***	−0.005	452
	(0.010)	(0.012)	(0.012)	
Source-text years from 1980, year FE	0.039***	0.038***	−0.005	439
	(0.010)	(0.012)	(0.012)	

Notes: Dependent variable is PCSI; public R1 is the omitted type. "Full controls" adds land-grant, medical-school, and urbanicity indicators and region fixed effects. University-mean rows use HC3 standard errors; the published row uses 150 universities for the type-only column and 144 with full controls. Era-adjusted means subtract each calendar year's cross-sectional mean from the policy-in-force panel before averaging within university. Source-text rows use university-years with directly available policy text, include source-year fixed effects, and cluster standard errors by university. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$.

E Transparent Linguistic Diagnostics

The diagnostics interpret the supervised measure; they are not independent validation of reader perceptions or alternative definitions of stance. Let $z(\cdot)$ denote a standardized policy-level component. The indices are

$$\begin{aligned}\text{Tone} = \tfrac{1}{6}\big[&z(\text{supportive}) + z(\text{second person}) + z(\text{inclusive we}) \\ &- z(\text{restrictive}) - z(\text{sanction}) - z(\text{obligation modal share})\big],\end{aligned} \tag{3}$$

$$\text{Clarity} = \tfrac{1}{3}\big[-z(\text{mean sentence length}) - z(\text{long-sentence share}) + z(\text{procedural share})\big], \tag{4}$$

$$\text{Legal Load} = \tfrac{1}{2}\big[z(\text{legalese}) + z(\text{IP technical})\big]. \tag{5}$$

Supportive, restrictive, sanction, procedural, legalese, and IP-technical components are constructed from transparent dictionaries implemented in the replication code; second-person and inclusive-first-person components capture direct and collective interpersonal language. Components are

normalized for policy length where appropriate and standardized before the composites are formed. The three diagnostics are empirically distinct: Tone and Clarity correlate at 0.143, Tone and Legal Load at 0.029, and Clarity and Legal Load at -0.296. Table E1 shows that adding log word count or log sentence count leaves the Tone relationship almost unchanged while reducing the Legal Load coefficient. In the 480 source-text years, the three-index model gives coefficients of 0.038 for Tone, 0.013 for Clarity, and 0.012 for Legal Load, with $R^2 = 0.151$.

Table E1: Linguistic Decomposition with Policy-Length Controls

	Baseline	+ Log words	+ Log sentences
Tone Index	0.062***	0.059***	0.059***
	(0.012)	(0.012)	(0.012)
Clarity Index	0.006	0.006	0.008
	(0.008)	(0.008)	(0.009)
Legal Load Index	0.013**	0.010	0.010
	(0.006)	(0.007)	(0.007)
Policy-length control	No	Log words	Log sentences
R^2	0.236	0.244	0.241
Observations	4,277	4,277	4,277

Notes: University-clustered standard errors in parentheses. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$.

F Temporal Robustness

The raw annual series can change because policy language changes or because the set of represented universities changes. Table F1 holds composition fixed using universities already represented by selected start years; every cohort has a lower mean in 2025 than at the start, and every university fixed-effects trend is negative. Table F2 removes annual carry-forward and uses only university-years with source policy text. The 2025 cross-section is not driven by unusually old policies: among the 70 universities whose policy in force in 2025 is at most five years old, 91 percent lie below the midpoint, against 92 percent of all 150.

Table F1: PCSI Trends Within Fixed University Cohorts

Cohort start	Universities	Start mean	2025 mean	FE trend per year	p
1990	34	0.461	0.421	-0.00109	0.019
1995	53	0.454	0.422	-0.00131	0.002
2000	93	0.447	0.422	-0.00112	0.008
2005	114	0.439	0.420	-0.00103	0.011

Notes: Each cohort contains the universities already represented in the policy-in-force panel by the start year. Trends come from regressions of PCSI on year with university fixed effects; standard errors are clustered by university.

Table F2: Temporal Trends Using Only Years with Source Policy Text

Start year	N	Pooled trend	FE trend	SE	p
1980	467	−0.00115	−0.00117	(0.00039)	0.003
1990	443	−0.00121	−0.00128	(0.00053)	0.016
2000	367	−0.00109	−0.00150	(0.00093)	0.105

Notes: University-years with directly available policy text from the start year onward. Standard errors are clustered by university.

G Clause Topics and Rule Forms

Clause topics. Sentences are assigned to topics by case-insensitive keyword rules: ownership and assignment (*assign*, *ownership*, *title*, *vest*, *belong*); disclosure (*disclos*, *invention report*); revenue sharing (*royalt*, *revenue*, *net income/proceeds*, *distribut*, *equity*); licensing and commercialization (*licens*, *commerciali*, *technology transfer*, *start-up*, *patent application*); sponsored research and third parties (*sponsor*, *grant*, *federal*, *government*, *contract*, *third part*); copyright and software (*copyright*, *software*, *courseware*, *scholarly work*); administration and disputes (*appeal*, *dispute*, *committee*, *review*, *waive*, *grievance*, *amend*); and definitions and scope (*means*, *defined*, *applies to*, *purpose*, *scope*). A sentence can belong to several topics; for the shift-share each sentence receives one primary topic, assigned in the order listed. Among sentences from 1980 onward, 59 percent match at least one topic. Ownership and assignment sentences are the most restrictive (mean score 0.335), and definitional and revenue-sharing sentences the least (0.448 and 0.453). Table G1 reports within-topic trends.

Shift-share. The decomposition uses the 63 universities with source text in both 1980–1999 and 2010–2025 and weights each university equally: category shares and category means are computed within university and period and then averaged. Writing $\bar{w}_k$ and $\bar{m}_k$ for the average share and mean of category k across the two periods and $\bar{m}$ for the share-weighted grand mean, the change in mean sentence score is decomposed into $\sum_k \bar{w}_k \Delta m_k$ (within categories) and $\sum_k (\bar{m}_k - \bar{m}) \Delta w_k$ (composition).

Rule forms. Following Crawford and Ostrom (1995), each sentence receives one deontic operator, assigned with priority prohibition (¬ *shall*, *must*, *may*, *will*, *can*; *cannot*; *prohibit*; *not permitted*) $>$ obligation (*shall*, *must*, *is required*, *requires*, *obligat-*, *agrees to*, *is expected to*) $>$ permission (*may*, *permit*, *entitled*, *discretion*, *can*, *allow*) $>$ none. Sanction language (*violat-*, *disciplin-*, *sanction*, *forfeit*, *terminat-*, *breach*, *penalt-*, *dismiss*, *revok-*) is flagged separately. Table G2 reports stance by form. At the policy level, the shares of obligation, prohibition, permission, and sanction sentences together explain 11 percent of the variation in PCSI across the 480 source-text university-years.

Table G1: Within-University Trends in Sentence Stance by Clause Topic, 1980–2025

Topic	Share	Mean score	FE trend per year	SE	p
Ownership & assignment	0.142	0.335	−0.00031	(0.00067)	0.644
Disclosure	0.057	0.357	−0.00223	(0.00107)	0.037
Revenue sharing	0.131	0.453	−0.00169	(0.00065)	0.010
Licensing & commercialization	0.133	0.432	−0.00129	(0.00085)	0.129
Sponsored research & third parties	0.131	0.341	−0.00092	(0.00062)	0.140
Copyright & software	0.092	0.412	−0.00111	(0.00079)	0.159
Administration & disputes	0.141	0.425	−0.00134	(0.00057)	0.018
Definitions & scope	0.083	0.448	−0.00112	(0.00085)	0.185
Other / general	0.409	0.427	−0.00070	(0.00057)	0.213
All sentences (= PCSI)	1.000		−0.00117	(0.00039)	0.002

Notes: Each row regresses the university-year mean sentence score within a topic on year and university fixed effects, using source-text university-years from 1980 and universities observed in at least two such years; standard errors are clustered by university. Topics are assigned by keyword rules and can overlap; share is the fraction of 1980–2025 sentences matching the topic. Shift-share for the 63 universities with source text in both 1980–1999 and 2010–2025: within-topic change −0.028, change in topic composition −0.003.

Table G2: Communicative Stance by Rule Form

Rule form	Share	Mean score	Restrictive	Supportive
Obligation	0.314	0.329	0.466	0.105
Prohibition	0.072	0.430	0.470	0.316
Permission	0.138	0.431	0.406	0.246
No deontic operator	0.475	0.467	0.226	0.160
Sanction language (any form)	0.037	0.334	0.487	0.124

Notes: Deontic operators follow the Institutional Grammar of Crawford and Ostrom (1995) and are assigned by keyword rules with priority prohibition > obligation > permission (Appendix G). Sanction language (the grammar's "or else" component) can co-occur with any form. Restrictive and supportive shares use the classifier's most likely label. Rule form explains 4.8 percent of sentence-level variance in stance. Shift-share across the eight form-by-sanction cells, 1980–1999 to 2010–2025: within-form change −0.030, change in form composition 0.002.

H Prediction Uncertainty and Alternative Constructions

PCSI aggregates calibrated class probabilities rather than hard labels, which preserves the classifier's uncertainty, but downstream regressions still treat the predicted scores as data. Without gold-standard labels linked to the analysis sample, design-based corrections (Egami et al., 2023; Battaglia et al., 2024) are not available, so we examine whether the headline results depend on uncertain predictions. Table H1 recomputes them from hard labels, from the 66 percent of sentences whose largest class probability is at least 0.9, and from the measure of framing net of rule form described in Section 2.5. Every construction places most policies below the midpoint, attributes about half of source-text variation to universities, shows a negative within-university trend, and yields a positive and significant private-R1 coefficient. Model confidence is also stable over time: from the 1970s onward, the mean largest class probability lies between 0.874 and 0.886 in every decade, so older text is not scored with systematically more uncertainty.

Table H1: Headline Results Under Alternative Constructions of PCSI

Construction	r with baseline	Mean	Share < 0.50	Between share	FE trend, 1980+	Private R1
Baseline PCSI		0.429	0.896	0.487	−0.00117***	0.040***
					(0.00039)	(0.012)
Hard labels	0.980	0.425	0.856	0.487	−0.00139***	0.042***
					(0.00044)	(0.013)
High-confidence sentences	0.952	0.418	0.875	0.495	−0.00134***	0.049***
					(0.00051)	(0.014)
Framing net of rule form	0.943	0.427	0.894	0.492	−0.00135***	0.032**
					(0.00037)	(0.013)

Notes: All columns use the 480 university-years with directly available policy text. Hard labels score each sentence 0, 0.5, or 1 by its most likely class. High-confidence uses only sentences whose largest calibrated class probability is at least 0.9 (about two-thirds of sentences). Framing net of rule form subtracts the mean score of each sentence's deontic-by-sanction cell before aggregation. The trend is the coefficient on year with university fixed effects, using universities with at least two source-text years from 1980; the private-R1 coefficient is from the Column (4) specification of Table 6. Standard errors clustered by university. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$.